Re-evaluating CIRAG on Vietnamese Multi-hop QA: A Refusal-Based Stopping Test and an Evidence-Based Replacement

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Abstract

CIRAG (Wei et al., 2026) compresses retrieved passages into (s, r, o) triples to reduce noise, and its authors report a large gain over earlier iterative retrieval methods. We reimplement the method and test it on the dataset and backbone used in the original paper, reproducing the reported margin to within 0.6 EM points. On ViMQA (Le et al., 2022), a Vietnamese multi-hop QA dataset, this margin does not appear. Four controlled experiments show that the cause lies neither in our code, the Vietnamese prompts, a broken pipeline stage, nor backbone capability. Instead, the cause is the method's stopping test. ACMG widens the context only when the model itself says it cannot answer, so it trusts the model's judgement rather than checking the context. As the backbone becomes stronger, it says this less often. The system therefore almost always stops at the narrowest context level and answers wrongly from there. We replace this test with one that checks the evidence itself. The new test requires the core triple set to span as many documents as a multi-hop question needs, and it raises EM from 50.80 to 60.40 with no extra model calls.

1 Introduction

Retrieval-augmented generation (Lewis et al., 2020) works well on simple questions but struggles with multi-hop reasoning because a single retrieval pass cannot gather pieces of evidence that only make sense together. Iterative retrieval (Trivedi et al., 2023) addresses this problem by dividing the search into several rounds. Each round uses what the system has just learned to form the next query. More recent work uses triples rather than passages as the unit of retrieval (Fang et al., 2024, 2025), which narrows the evidence and reduces noise.

CIRAG (Wei et al., 2026) addresses two separate problems in this line of work. The first is greedy single-path expansion. Earlier methods keep only the single best triple at each step, so an early mistake affects later steps, while the parallel evidence branches needed for complex questions are lost. The second problem is a mismatch between the detail provided by the evidence and the detail the question needs. Triples are clean and low-noise but lack context, while passages provide full context with more noise. Thus, no fixed level suits every question.

To limit greedy expansion, CIRAG proposes the ICI retrieval module (Iterative Construction-Integration), based on the Construction-Integration model from cognitive psychology (Kintsch and Van Dijk, 1978). The construction stage collects every triple from the newly retrieved passages and ranks them to form a broad candidate pool. The integration stage gives this pool and the reasoning history to the backbone model. The model selects the triples that the question needs, called the core triple set, and writes the query for the next round. Because the model can select several triples at once, the core set can include evidence from several reasoning branches rather than only one.

To address the level-of-detail problem, CIRAG proposes the ACMG generation module (Adaptive Cascaded Multi-Granularity Knowledge-Enhanced Generation). At each level, the system gives the model the context from that level and asks it to answer. If the context is insufficient, the model must return a fixed refusal string instead of guessing. The module begins at the triple level and moves to sentences and then passages only after it receives a refusal.

We reimplement CIRAG and evaluate it on ViMQA (Le et al., 2022), a Vietnamese multi-hop QA dataset. On the original paper's dataset, our code reproduces its reported gain, but this gain does not appear on ViMQA. The rest of this report investigates why. We make four contributions:

- A verified reimplementation. We run it on

2WikiMultihopQA with the original backbone, and the margin over IRCoT it measures sits within 0.6 EM points of the one the paper reports (§6).

- CIRAG relies on the model's refusal to tell whether the context is enough. That signal is wrong in both directions: the model refuses even when the evidence is already there, and answers even when it is missing (§7).
- A replacement stopping test that asks the core triple set to span at least two documents instead of waiting for the model to refuse. It lifts EM from 50.80 to 60.40 at no extra model calls (§8).
- Four explanations for why CIRAG's margin does not appear on ViMQA, each ruled out by an experiment that changes exactly one variable (§9).

2 Related Work

Text-based iterative retrieval. IRCoT (Trivedi et al., 2023) alternates between writing a reasoning step and retrieving evidence, using the sentence it has just written as the next query. Self-Ask (Press et al., 2023) divides a question into explicit subquestions. Because each round adds a whole passage to the context, noise increases over successive rounds.

Triple-based iterative retrieval. To reduce this noise, one line of work uses triples instead of passages as the retrieval unit. HippoRAG (Jimenez Gutierrez et al., 2024) builds a knowledge graph from extracted triples, while KiRAG (Fang et al., 2025) extends a chain of triples round by round. However, these methods keep only the best triple at each step and fix the representation at the triple level.

CIRAG differs from both lines of work in exactly these two ways. It keeps a set of core triples rather than a single chain, and it widens the context in stages rather than using a fixed level.

3 Method

CIRAG runs two modules in sequence (Figure 1). ICI retrieves and filters evidence into a core triple set, and ACMG produces the answer from that set. The original paper includes a third module, Trajectory Distillation, which trains a small model to copy the reasoning steps of a large model and thus reduce cost. We do not implement this module, so every result in this report is compared with the paper's *w/o TD* setting.

ICI. Each round has two stages. The construction stage retrieves the top-K passages for query q_t, collects their triples, and reranks them to select the top-N candidates. The integration stage gives the candidate pool and the history $H_{<t}$ to the backbone model. The model keeps the core triple set F_t and writes the next query q_{t+1}. The loop stops when the model emits `<no question>`, after at most L rounds.

Each triple records its source sentence and passage. Two reverse maps, written M^S and M^D, turn F into the sentence set $S = M^S(F)$ and the passage set $D = M^D(F)$. The three sets F, S, D are the context levels used by ACMG.

ACMG. The system offers F, then S, then D to the model and asks for an answer, following the order triples $\prec$ sentences $\prec$ passages. If the context is not enough, the model must return a fixed refusal string. The first level that is not refused becomes the answer:

$$g^\star = \min_{\prec}\{g \in G : \mathrm{Suf}(a^{(g)}) = 1\}, \quad a^\star = a^{(g^\star)} \tag{1}$$

where $\mathrm{Suf}(a)$ is true when a does not match the refusal template. If no level qualifies, the system falls back to the passage-level answer. Both the Suf test and the fact that S and D derive from F are the source of the problems in §7.

4 Data

ViMQA (Le et al., 2022) is a Vietnamese multihop QA dataset written by hand from Vietnamese Wikipedia articles and based on the HotpotQA layout (Yang et al., 2018). Each question includes ten passages. Two contain the evidence, while eight act as distractors. The dataset also provides labels that mark the sentence containing the evidence.

We randomly sample 500 questions from the validation split. If each question searched only its own ten passages, the system would almost always find both evidence passages, making retrieval unimportant. Following IRCoT, we merge the passages from all questions and remove duplicates by title, producing a shared corpus of 1,969 passages. The system must then find two evidence passages among nearly two thousand rather than among ten.

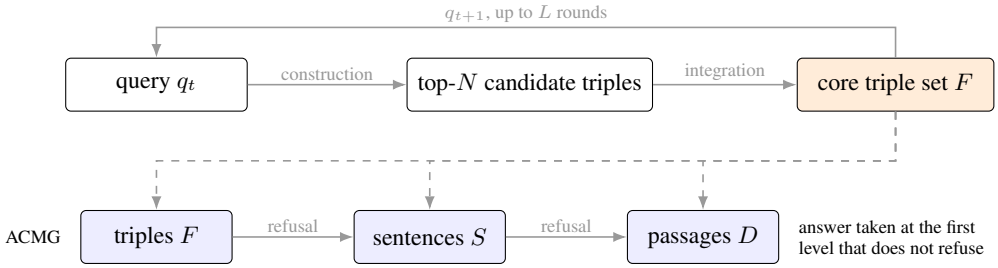


Figure 1: The CIRAG pipeline. ICI alternates between the construction and integration stages to build the core triple set F. ACMG first tries to answer from F and only moves to sentences and then passages when the model returns a refusal. The dashed arrows show that all three levels derive from F, so a gap left by the integration stage carries into all three (§7.2).

5 Experimental Setup

The hyperparameters follow the original paper (K=10, N=30, L=4). We replace two components to support Vietnamese. First, we use the Vietnamese_Embedding model, which achieves recall@10 of 0.920 on the 1,969-passage corpus. Second, we rewrite the triple extraction prompt in Vietnamese, producing 22,697 triples for the full corpus. The reasoning model is Gemini 2.5 Flash, while triple extraction uses the cheaper Flash Lite because this stage is easier and accounts for most calls.

Metrics. Let $\hat{a}$ be the system answer, a the reference answer, and $\nu(\cdot)$ a normalisation function. EM is 1 only when $\nu(\hat{a})$ and $\nu(a)$ match exactly. F1 is the harmonic mean of precision and recall over the tokens shared by the two answers. F1 ignores token order, while EM also requires the order to match.

Vietnamese requires two adjustments to ν. First, the same accented letter can be stored as two byte strings that look identical on screen, so ν first maps every string to one Unicode normal form. Second, a Vietnamese word can contain several syllables ("học sinh" is one word but two), and F1 varies with the split. We therefore report both.

Statistical significance. Two methods answer the same set of questions, so for most questions they are either both right or both wrong. Only questions that one method answers correctly and the other answers wrongly distinguish them. The McNemar test counts exactly this group. If the methods were equally good, the counts favouring each method should be similar. A small p means that the observed split is unlikely to be due to chance.

EM gives 0 to answers that are right in content. EM assigns a score only when the two strings are identical, so any length mismatch between the system answer and the reference becomes a 0. This mismatch occurs in two directions. First, the reference answer can be long. The system names every part but reverses the order, giving F1 of 1.00 and EM of 0. Among the 99 questions with a reference longer than five syllables, only 10 reach EM 1. This direction cannot be repaired, so it sets a ceiling on every EM result in this report. Second, the system answer can be long. Models answering in Vietnamese use full sentences, while the reference is a short phrase. We address this case by trimming the answer before scoring.

What we hold fixed. We compare CIRAG with two baselines. NativeRAG retrieves once and answers, while IRCoT retrieves passages iteratively. All three methods share the retrieval module, corpus, and backbone model. Thus, they differ only in how they select evidence and write their answers. The trimming step also applies to all three. If we trimmed only CIRAG, the results would show which method writes shorter answers rather than which one reasons better.

6 Results

Our code reproduces the original paper. The ViMQA results are meaningful only if our code is correct, so we first run it on 2WikiMultihopQA with Qwen2.5-7B, the dataset and backbone used in the original paper. CIRAG beats IRCoT by 14.0 EM points, compared with the 13.4 points reported in the paper (Table 1). Each individual result is within 4.2 points of theirs. Some difference is expected because we run 100 questions on a 770-passage corpus, which makes retrieval easier than in their setup. The margin between the two methods avoids this effect because both use the same corpus, and it differs by only 0.6 points.

Method	EM	F1	Δ	p
2WikiMultihopQA, Qwen2.5-7B, 100 questions				
IRCoT (ours)	40.00	44.63		
CIRAG (ours)	54.00	57.97	**+14.0**	0.013
IRCoT (paper)	36.4	45.7		
CIRAG *w/o TD* (paper)	49.8	59.7	**+13.4**	
ViMQA, Gemini 2.5 Flash, 500 questions				
NativeRAG	55.20	67.95	−3.8	0.056
IRCoT	59.00	73.27		
CIRAG, original test	50.80	66.66	−8.2	<0.001
+ fallback level	57.00	73.05	−2.0	0.29
+ new stopping test	**60.40**	**75.08**	**+1.4**	0.44

Table 1: Δ is the EM difference against IRCoT and p is the McNemar test for that difference (the paper reports no p). The last three rows are our own build: the original stopping test of Equation 1, then the two changes of §8 added one at a time.

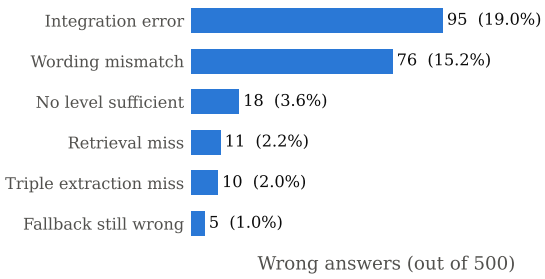


Figure 2: The 215 wrong answers, grouped by the stage where the error first appears.

The margin does not appear on ViMQA. NativeRAG, which retrieves only once, falls 3.8 EM points behind IRCoT. This gap confirms that ViMQA does require several retrieval rounds, so comparing two iterative methods is meaningful. CIRAG falls 2.0 EM points behind IRCoT. However, CIRAG is right and IRCoT is wrong on only 31 questions, compared with 41 in the other direction, so the gap is not solid evidence ($p = 0.29$). We can therefore say that CIRAG *no longer leads* IRCoT as it did on 2WikiMQA, but not that CIRAG *trails* it.

7 Error Analysis

Figure 2 groups the 215 wrong answers by the first stage at which each one goes wrong. Retrieval and triple extraction together account for only 21, so evidence gathering is not the bottleneck. Instead, the bottleneck is the integration stage, where the model selects the core triple set. This stage alone accounts for 95. A further 76 answers are correct in content but receive 0 on EM because their wording differs (§5). This is a problem with the metric rather than the method. The next two subsections examine the 95 integration errors.

Configuration	EM	F1
Full cascade	57.00	73.05
triples only	57.40	72.99
sentences only	60.80	**75.38**
passages only	**61.20**	75.16
triples + sentences	57.00	73.03
triples + passages	56.80	72.78
$L = 1$	52.60	67.52
$L = 2$	57.40	73.18

Table 2: Ablation. Every variant keeps the fallback level from §8, so the rows compare against each other but not directly against Table 3 of the original paper. The last two rows show performance levelling off from L=2 onward.

7.1 The stopping test fires too early

The cascade provides no benefit in our setting and sometimes causes harm. Two variants that remove it, one that always reads sentences and one that always reads passages, beat the full cascade by 3.8 and 4.2 EM points ($p < 0.005$, McNemar). Only the triples-only variant matches the cascade. Table 2 therefore contrasts with Table 3 of the original paper, where the cascade beats every single-level variant.

The cascade fails because it relies on a signal that a strong backbone does not send. The Suf test permits a wider context only when the model returns the refusal string, but Gemini 2.5 Flash rarely returns it. For 361 of the 500 questions, the system stops at the triple level immediately. §8 explains this mechanism and proposes a replacement.

Table 2 also shows that performance becomes flat from L=2 onward, so the number of rounds is not the bottleneck.

7.2 Every context level contains the filter's gaps

The second failure occurs when the system refuses to answer even though the evidence has already been retrieved. All three levels come from the same core triple set. Thus, anything dropped by the filter is missing from F, S and D alike, and widening the context only presents the same gap in a longer form. The model refuses at every level, and the system enters the default branch of Equation 1. This branch handles 73 questions, all incorrect, although 66 of them have retrieval recall of 1.0. The retrieval module found enough evidence, but

the filter removed it. The group that suffers most is comparison questions, 24 out of 51, because the filter usually keeps triples about only one of the two entities.

8 Replacing the Stopping Test with an Evidence Test

Suf uses the refusal string as its signal of whether the context is sufficient, but this signal is wrong in both directions. The model refuses when the evidence is already present (§7.2), and it answers when evidence is missing (§7.1). We address the first direction first because the gains from the two changes are additive.

The model refuses although the evidence is present. We add a final level that is not built from the core set. When no level in the cascade produces an answer, the system rereads all retrieved passages instead of only those indicated by the triples. This level raises EM from 50.80 to 57.00 and reduces refusals from 73 to 35. It handles 38 questions, of which 31 are answered correctly.

The model answers although the evidence is missing. In this case, the system stops at the triple level with an answer that reads well but is wrong, and the output gives no warning. The model is not inventing answers. Among the 262 questions whose answers can be string-matched, 252 contain the answer word for word in the core triple set. The model reads the triples it receives correctly, but the triples themselves are insufficient.

We can measure which sets are sufficient by counting how many passages each set spans, read off the provenance map M^D (§3). Sets that span a single passage score 35.7 EM, while those that span two score 71.6. Every ViMQA question requires evidence from at least two passages. A core set drawn from one passage is therefore insufficient by construction, however willing the model is to answer.

We therefore add a test on the evidence alongside the existing test on the answer:

$$\mathrm{Suf}'(a, F) = \mathrm{Suf}(a) \cdot \mathbf{1}\left[|M^D(F)| \geq h\right] \quad (2)$$

where h is the number of documents the question requires. When the test fails, the system skips the narrow levels and reads the widest one directly. The test only rereads information the method already stores, so it costs no extra model calls, and $h = 2$ comes from the known structure of the dataset rather than from tuning on the evaluation set.

Results. With both changes, CIRAG moves from behind IRCoT to ahead of it (Table 1). The new test alone adds 3.40 points, with a paired McNemar test at $p = 0.0005$. It fixes 20 questions while breaking only 3. When we change the backbone to Qwen3-30B-A3B, the improvement has the same direction but is smaller and not significant (55.40 to 57.00 EM, $p = 0.12$). Notably, after the change, CIRAG leads IRCoT by exactly 1.4 points on both backbones, while it previously matched or trailed IRCoT.

What the change does not achieve. It does not beat the passages-only variant in Table 2: 60.40 against 61.20, with McNemar at $p = 0.56$. The remaining benefit is lower cost, and this amounts to only 2.3% of total input tokens because the ICI stage, rather than the answering step, consumes most of them.

9 Ruling Out Four Explanations

We test each explanation below with an experiment that changes exactly one variable. The reproduction in §6 already rules out the first explanation, a faulty reimplementation.

For the second explanation, we wrote the Vietnamese prompts ourselves, while the English prompts came from the authors. Prompt quality could therefore explain the gap. We rerun all 500 questions with the original prompts and add one line requiring a Vietnamese answer. The result is 57.40 EM against 57.00, while the distribution across context levels changes very little.

The third explanation asks whether one pipeline stage fails on the Vietnamese data. We keep the backbone fixed and compare retrieval recall, the share of integration steps that keep at least one triple, and the share that produce a next query. All three measures differ by less than five points across the two datasets, so the pipeline behaves similarly in both languages.

The fourth explanation requires a correction. If NativeRAG is the reference rather than IRCoT, the CIRAG margin shrinks from 8.8 to 1.8 points as the backbone becomes stronger. This can easily be read as evidence that the backbone matters. However, NativeRAG benefits unusually strongly from a better backbone because it reads all ten passages directly. The shrinking margin therefore reflects improvement in NativeRAG rather than decline in CIRAG. If IRCoT is the reference, changing the backbone alters the margin by only 0.8 points, com-

pared with the 15.2 points caused by changing the dataset.

What remains. After ruling out all four explanations, the only significant difference is by question type. CIRAG leads IRCoT by 4.5 points on yes/no questions ($n = 156$, $p = 0.15$) but trails by 4.9 points on open-ended questions ($n = 344$, $p = 0.031$). Open questions require the system to reproduce a specific phrase word for word. Compressing context into triples has the greatest cost in such cases because triples have already removed the original wording.

10 Conclusion

CIRAG's 14-point margin does not appear on ViMQA, and the cause lies in the method's own stopping test. The lesson extends beyond CIRAG. A stopping test based on a model's behaviour can be trusted only while the model continues to show that behaviour. As the backbone becomes stronger, it refuses less often, so wider context levels remain unused without any signal in the output. A test based on the evidence prevents this silent failure.

Limitations

Our main conclusion rests on two datasets, and the reproduction on 2WikiMQA covers only 100 questions with a 770-passage corpus, far fewer than the 1,000 questions used in the original paper. A smaller setting makes retrieval easier, so the reproduction only works in one direction: it removes the suspicion of faulty code, but it does not prove the two builds are equivalent.

Ruling out the backbone rests on two models, enough to show its contribution is small next to the dataset effect, but not enough to describe how the benefit varies with capability. Finally, the two ways EM scores 0 on a correct answer (§5) affect all three methods: the 99 questions with long reference answers alone account for 89 of CIRAG's 215 errors.

References

Jinyuan Fang, Zaiqiao Meng, and Craig MacDonald. 2024. TRACE the evidence: Constructing knowledge-grounded reasoning chains for retrieval-augmented generation. In *Findings of the Association for Computational Linguistics: EMNLP*, pages 8472–8494.

Jinyuan Fang, Zaiqiao Meng, and Craig Macdonald. 2025. KiRAG: Knowledge-driven iterative retriever for enhancing retrieval-augmented generation. *arXiv preprint arXiv:2502.18397*.

Bernal Jimenez Gutierrez, Yiheng Shu, Yu Gu, Michihiro Yasunaga, and Yu Su. 2024. HippoRAG: Neurobiologically inspired long-term memory for large language models. In *Advances in Neural Information Processing Systems*, volume 37, pages 59532–59569.

Walter Kintsch and Teun A. Van Dijk. 1978. Toward a model of text comprehension and production. *Psychological Review*, 85(5):363.

Khang Le, Hien Nguyen, Tung Le Thanh, and Minh Nguyen. 2022. VIMQA: A Vietnamese dataset for advanced reasoning and explainable multi-hop question answering. In *Proceedings of the Thirteenth Language Resources and Evaluation Conference*, pages 6521–6529. European Language Resources Association.

Patrick Lewis, Ethan Perez, Aleksandra Piktus, Fabio Petroni, Vladimir Karpukhin, Naman Goyal, Heinrich Küttler, Mike Lewis, Wen-tau Yih, Tim Rocktäschel, et al. 2020. Retrieval-augmented generation for knowledge-intensive NLP tasks. *Advances in Neural Information Processing Systems*, 33:9459–9474.

Ofir Press, Muru Zhang, Sewon Min, Ludwig Schmidt, Noah A. Smith, and Mike Lewis. 2023. Measuring and narrowing the compositionality gap in language models. In *Findings of the Association for Computational Linguistics: EMNLP 2023*, pages 5687–5711.

Harsh Trivedi, Niranjan Balasubramanian, Tushar Khot, and Ashish Sabharwal. 2023. Interleaving retrieval with chain-of-thought reasoning for knowledge-intensive multi-step questions. In *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics*, pages 10014–10037.

Zili Wei, Xiaocui Yang, Yilin Wang, Zihan Wang, Weidong Bao, Shi Feng, Daling Wang, and Yifei Zhang. 2026. CIRAG: Construction-integration retrieval and adaptive generation for multi-hop question answering. *arXiv preprint arXiv:2601.06799*.

Zhilin Yang, Peng Qi, Saizheng Zhang, Yoshua Bengio, William W. Cohen, Ruslan Salakhutdinov, and Christopher D. Manning. 2018. HotpotQA: A dataset for diverse, explainable multi-hop question answering. In *Proceedings of the 2018 Conference on Empirical Methods in Natural Language Processing*, pages 2369–2380.